

4.1 Vector Spaces and Subspaces

I

Definition

A vector space is a nonempty set V of objects, called vectors, on which are defined two operations, multiplication by scalars (real numbers) and addition, subject to the ten axioms listed below. The axioms must hold for all vectors $\vec{u}, \vec{v}$ and $\vec{w}$ in V and for all scalars c, d .

1. The sum of $\vec{u}$ and $\vec{v}$, $\vec{u} + \vec{v}$, is in V .
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$.
4. There is a zero vector in V such that $\vec{u} + \vec{0} = \vec{u}$.
5. For each $\vec{u}$ in V , there is a vector $-\vec{u}$ in V such that $\vec{u} + (-\vec{u}) = \vec{0}$.
6. The scalar multiple of $\vec{u}$ by c , $c\vec{u}$, is in V .
7. $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$.
8. $(c+d)\vec{u} = c\vec{u} + d\vec{u}$.
9. $c(d\vec{u}) = (cd)\vec{u}$.
10. $1\vec{u} = \vec{u}$.

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subspaces

Definition

A subspace of a vector space V is a subset H of V that has three properties:

- a. The zero vector is in H .
- b. H is closed under vector addition. That is, for each $\vec{u}$ and $\vec{v}$ in H , the sum $\vec{u} + \vec{v}$ is in H .
- c. H is closed under scalar multiplication. That is, for each $\vec{u}$ in H and each scalar c , the vector $c\vec{u}$ is in H .

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A plane in $\mathbb{R}^3$ not through the origin is not a subspace of $\mathbb{R}^3$, because it does not contain the zero vector.

Similarly, a line in $\mathbb{R}^2$ not going through the origin is not a subspace of $\mathbb{R}^2$.

4.1 Vector Spaces and Subspaces (cont'd)

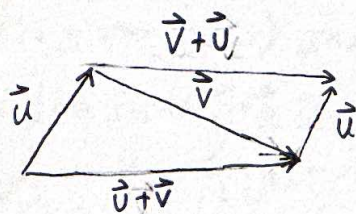
II

Example 1

The spaces $\mathbb{R}^n$, where $n \geq 1$, are the premier examples of vector spaces. The geometric intuition developed for $\mathbb{R}^3$ will help you understand and visualize many concepts throughout the chapter.

Example 2

Let V be the set of all arrows in $\mathbb{R}^3$, with two arrows regarded as equal if they have the same length and point in the same direction. Define addition geometrically by the parallelogram rule.



$$\vec{u} + \vec{v} = \vec{v} + \vec{u}$$

Example 3

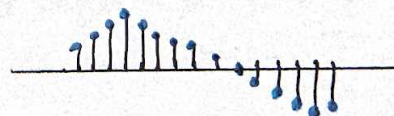
Let $\mathcal{S}$ be the space of all doubly infinite sequences of numbers (usually written in row

rather than a column).

$$\{y_k\} = (\dots, y_{-2}, y_{-1}, y_0, y_1, y_2, \dots)$$

If z_k is another element of $\mathcal{S}$, then the sum $\{y_k\} + \{z_k\}$ is the sequence $\{y_k + z_k\}$ formed by adding corresponding terms of $\{y_k\}$ and $\{z_k\}$. The scalar multiple $c\{y_k\}$ is the sequence $\{cy_k\}$. The vector space axioms are verified in the same way as for $\mathbb{R}^n$.

Elements of $\mathcal{S}$ arise in engineering, for example, whenever a signal is measured (or sampled) at discrete times. A signal might be electrical, mechanical, optical, biological, audio, and soon. The digital signal processors mentioned in the chapter introduction use discrete signals. For convenience, we will call $\mathcal{S}$ the space of (discrete-time) signals.



discrete-time signal example

4.1 Vector Spaces and Subspaces (cont'd)

III

Example 4

For $n \geq 0$, the set $\mathbb{P}_n$ of polynomials of degree at most n consists of all polynomials of the form

$$\vec{p}(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n \quad (4)$$

where the coefficients $a_0, \dots, a_n$ and the variable t are real numbers. The degree of $\vec{p}$ is the highest power of t in (4) whose coefficient is not zero. If $\vec{p}(t) = a_0 \neq 0$, the zero polynomial is included in $\mathbb{P}_n$ even though its degree, for technical reasons, is not defined.

If $\vec{p}$ is given by (4) and if $\vec{q}(t) = b_0 + b_1 t + \cdots + b_n t^n$, then the sum $\vec{p} + \vec{q}$ is defined by

$$\begin{aligned} (\vec{p} + \vec{q})(t) &= \vec{p}(t) + \vec{q}(t) \\ &= (a_0 + b_0) + (a_1 + b_1)t + \cdots + (a_n + b_n)t^n \end{aligned}$$

The scalar multiple $c\vec{p}$ is the polynomial defined by

$$(c\vec{p})(t) = c\vec{p}(t) = ca_0 + (ca_1)t + \cdots + (ca_n)t^n$$

Example 5

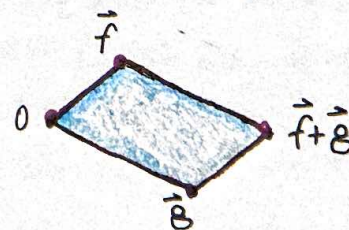
Let V be the set of all real-valued functions defined on a set $\mathbb{D}$. (Typically $\mathbb{D}$ is the set of real numbers or some interval on the real line). Functions are added in the usual way: $\vec{f} + \vec{g}$ is the function whose value at t in the domain $\mathbb{D}$ is $\vec{f}(t) + \vec{g}(t)$. Likewise, for a scalar c and an $\vec{f}$ in V , the scalar multiple $c\vec{f}$ is the function whose value at t is $c\vec{f}(t)$. For instance, if $\mathbb{D} = \mathbb{R}$, $\vec{f}(t) = 1 + \sin 2t$, and $\vec{g}(t) = 2 + \frac{1}{2}t$, then

$$(\vec{f} + \vec{g})(t) = 3 + \sin 2t + \frac{1}{2}t \quad \text{and}$$

$$(2\vec{g})(t) = 4 + t$$

Two functions in V are equal if and only if their values are equal for every t in $\mathbb{D}$. Hence the zero vector in V is the function that is identically zero, $\vec{f}(t) = 0$ for all t , and the negative of $\vec{f}$ is $(-1)\vec{f}$.

It is important to think of each function in the vector space V of ex. 5 as a single object, or a point or vector in the vector space.



4.1 Vector Spaces and Subspaces (cont'd)

IV

Subspaces

In many problems, a vector space consists of an appropriate subset of vectors from some larger vector space. In this case, only three of the ten vector space axioms need to be checked; the rest are automatically satisfied.

Definition

A subspace of a vector space V is a subspace H of V that has three properties:

- The zero vector of V is in H .
- H is closed under vector addition. That is, for each $\vec{u}$ and $\vec{v}$ in H , the sum $\vec{u} + \vec{v}$ is in H .
- H is closed under scalar multiplication. That is, for each $\vec{u}$ in H and each scalar c , the vector $c\vec{u}$ is in H .



A subspace of V .

Example 6

The set consisting of only the zero vector in a vector space V is a subspace of V , called the zero subspace and written as $\{\vec{0}\}$.

Example 7

Let $\mathbb{P}$ be the set of all polynomials with real coefficients, with operations in $\mathbb{P}$ defined as for functions. Then $\mathbb{P}$ is a subspace of the space of all real-valued functions defined in $\mathbb{R}$. Also, for each $n \geq 0$, $\mathbb{P}_n$ is a subspace of $\mathbb{P}$, because $\mathbb{P}_n$ is a subset of $\mathbb{P}$ that contains the zero polynomial, the sum of two polynomials in $\mathbb{P}_n$ is also in $\mathbb{P}_n$, and a scalar multiple of a polynomial in $\mathbb{P}_n$ is also in $\mathbb{P}_n$.

Example 8

The set of finitely supported signals $\mathcal{S}_f$ consists of the signals $\{y_k\}$, where only finitely many of the y_k are nonzero. Since the zero signal $\vec{0} = (\dots, 0, 0, 0, \dots)$ has no nonzero entries, it is clearly an element of $\mathcal{S}_f$. If two signals with finitely many nonzeros are added, the resulting signal will have finitely many nonzeros. Similarly if a signal with finitely many nonzeros are scaled, the result will still have finitely many nonzeros. Thus $\mathcal{S}_f$ is a subspace of $\mathcal{S}$, the discrete-time signals.

4.1 Vector Spaces and Subspaces (cont'd)

V

Example 9

The vector space $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$ because $\mathbb{R}^2$ is not even a subset of $\mathbb{R}^3$. (The vectors in $\mathbb{R}^3$ all have three entries, whereas the vectors in $\mathbb{R}^2$ only have two). The set

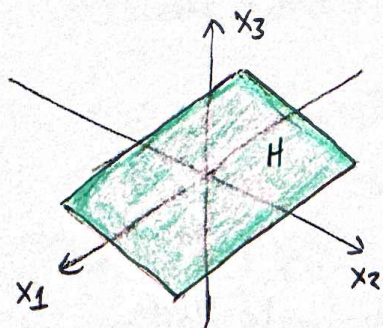
$$H = \left\{ \begin{bmatrix} s \\ t \\ 0 \end{bmatrix} : s \text{ and } t \text{ are real} \right\}$$

show that H is
a subspace
of $\mathbb{R}^3$

is a subset of $\mathbb{R}^3$ that "looks" and "acts" like $\mathbb{R}^2$, although it is logically distinct from $\mathbb{R}^2$.

Solution

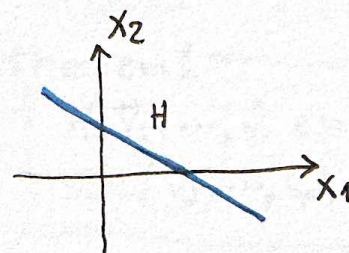
The zero vector is in H , and H is closed under vector addition and scalar multiplication because these operations on vectors in H always produce vectors whose third entries are zero. Thus H is a subspace of $\mathbb{R}^3$.



The x_1x_2 -plane
is a subspace
of $\mathbb{R}^3$.

Example 10

A plane in $\mathbb{R}^3$ not through the origin, is not a subspace of $\mathbb{R}^3$, because the plane does not contain the zero vector of $\mathbb{R}^3$. Similarly, a line in $\mathbb{R}^2$ not through the origin is not a subspace of $\mathbb{R}^2$.



H is not a subspace of $\mathbb{R}^2$.

4.1 Vector Spaces and Subspaces (cont'd)

VI

A subspace spanned by a set

Example 11

Given $\vec{v}_1$ and $\vec{v}_2$ in vector space V , let $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$. Show that H is a subspace of V .

Solution

The zero vector is in H , since $\vec{0} = 0\vec{v}_1 + 0\vec{v}_2$. To show that H is closed under vector addition, take two arbitrary vectors in H , say,

$$\vec{u} = s_1\vec{v}_1 + s_2\vec{v}_2 \quad \text{and} \quad \vec{w} = t_1\vec{v}_1 + t_2\vec{v}_2$$

By axioms 2, 3 and 8 for the vector space V ,

$$\begin{aligned} \vec{u} + \vec{w} &= (s_1\vec{v}_1 + s_2\vec{v}_2) + (t_1\vec{v}_1 + t_2\vec{v}_2) \\ &= (s_1 + t_1)\vec{v}_1 + (s_2 + t_2)\vec{v}_2 \end{aligned}$$

So $\vec{u} + \vec{w}$ is in H . Furthermore, if c is any scalar, then by axioms 7 and 9

$$\begin{aligned} c\vec{u} &= c(s_1\vec{v}_1 + s_2\vec{v}_2) \\ &= (cs_1)\vec{v}_1 + (cs_2)\vec{v}_2 \end{aligned}$$

which shows that $c\vec{u}$ is in H and H is closed under multiplication. H is a subspace of V .

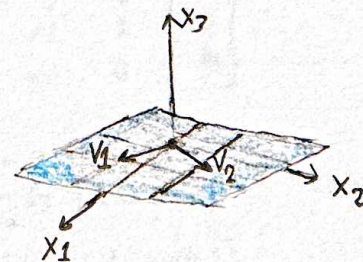


Figure: Example of a subspace.

Theorem 1

If $\vec{v}_1, \dots, \vec{v}_p$ are in a vector space V , then $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$ is a subspace of V .

Example 12

Let H be the set of all vectors of the form $(a-3b, b-a, a, b)$, where a and b are arbitrary scalars. That is, let

$$H = \{(a-3b, b-a, a, b) : a \text{ and } b \text{ in } \mathbb{R}\}.$$

Show that H is a subspace of $\mathbb{R}^4$.

Solution

Write the vectors in H as column vectors. Then an arbitrary vector in H has the form

∴

4.1 Vector Spaces and Subspaces (cont'd)

VII

$$\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow$ $\uparrow$
 $\vec{v}_1$ $\vec{v}_2$

This calculation shows that $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$, where $\vec{v}_1$ and $\vec{v}_2$ are the vectors indicated above. Thus H is a subspace of $\mathbb{R}^4$ by theorem 1.

Example 13

For what value(s) of h will y be in the subspace of $\mathbb{R}^3$ spanned by $\vec{v}_1, \vec{v}_2, \vec{v}_3$ if

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \text{ and } y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$

Solution

Let $H = \text{Span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$, for y to be in H , it must be a linear combination of $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$.

Set up an augmented matrix and row reduce

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ -1 & -4 & 1 & 3 \\ -2 & -7 & 0 & h \end{array} \right]$$

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1 \\ R_3 &\leftarrow R_3 + 2R_1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 3 & -6 & h-8 \end{array} \right]$$

$$R_3 \leftarrow R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & (h-8)+3 \end{array} \right]$$

Cleaning up

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & h-5 \end{array} \right]$$

We see that $\vec{y}$ is in H for exactly $h=5$.